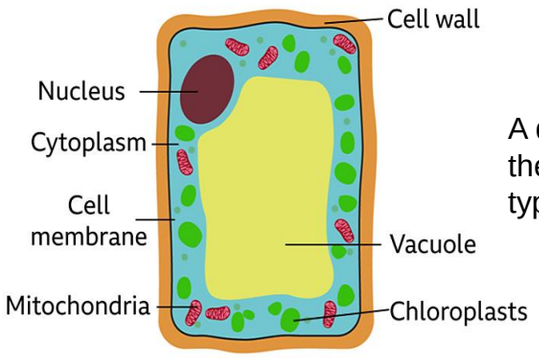
Knowledge Organiser
Year 7: Interdependence – Photosynthesis



A diagram showing the organelles in a typical plant cell

Animal vs. Plant Cells
- Cells are the smallest units of life and the building blocks of all organisms
- Each organelle (part of a cell) has its own function
- Animal and plant cells have some organelles in common e.g. cell membrane, nucleus and cytoplasm
- Plant cells have some unique organelles e.g. chloroplasts, permanent vacuole and cell wall

What is cellular respiration?
- All living cells must carry out cellular respiration
- This is a chemical reaction which releases the chemical energy stored in glucose into useful forms
- Most animals get glucose from the food they eat
- Plant cells also perform respiration, but they do not eat, so must get glucose through photosynthesis

GLUCOSE + OXYGEN ⟶ CARBON DIOXIDE + WATER

The chemical word equation for aerobic respiration

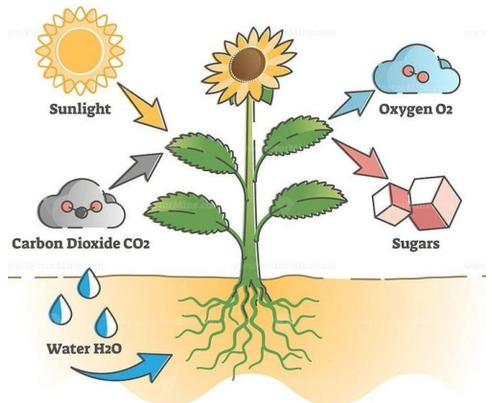
What is photosynthesis?
- Because plants cannot get the glucose they need from food, the must make their own
- Photosynthesis is a chemical reaction which uses light energy from the Sun to create glucose
- Glucose is then used by plants as a store of chemical energy which can be released through respiration

CARBON DIOXIDE + WATER ⟶ GLUCOSE + OXYGEN

The chemical word equation for photosynthesis

How does photosynthesis work?
- Photosynthesis requires **light energy**
- The organelles called **chloroplasts** absorb light energy using a green pigment called **chlorophyll**
- Light energy is used to react together **carbon dioxide** and **water** to create **glucose** and **oxygen**
- Light is not a **reactant** in this process because it is not used up

A diagram showing the reactants and products in photosynthesis



Adaptations for photosynthesis
- An adaptation is a variation (difference) which makes an organism better suited for a task
- Plants have many adaptations to make them more effective at photosynthesis

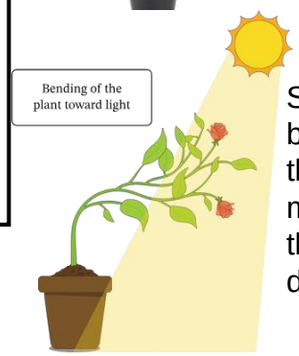
Plants will try to create **a large surface area** to absorb the maximum amount of sun light



Some plants grow a few large, spaced out leaves



Others grow many small leaves



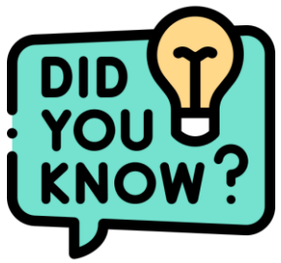
Some plants bend towards the Sun as it moves through the day



Others grow very tall stems to try and out compete other plants for Sunlight

Part	Function
Stem	Transport through the plant, keep it upright
Roots	Absorb water and minerals from the soil
Flower	Attract pollinating insects
Leaf	Absorb sunlight for photosynthesis
Tuber e.g. potato	Used to store nutrients like sugar

What factors affect the growth rate of plants?
- **Carbon dioxide concentration**: If this is too low plants cannot photosynthesise and produce glucose
- **Light intensity**: If this is too low plants cannot convert CO_2 and H_2O into glucose through photosynthesis
- **Temperature:** If temperature is too high plant cells are damaged, if it is too low then the rate of photosynthesis decreases
- **Concentration of chlorophyll:** Some plant diseases can reduce the amount of chlorophyll, which is needed to absorb Sunlight for photosynthesis

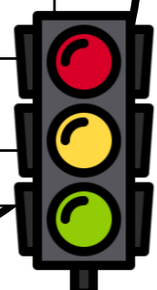


If black surfaces absorb the most radiation, **why are plants green**?

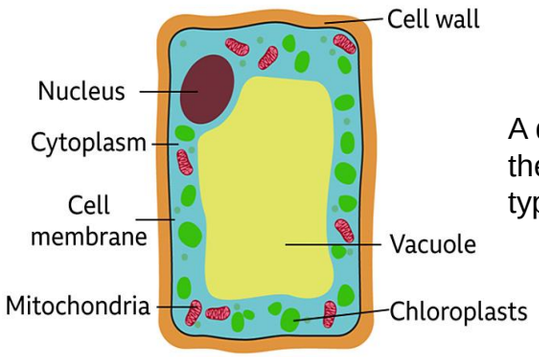
It stops the leaves getting too hot which would damage the cells.

RAG Yourself!

Learning Outcomes;	RAG
Describe the importance of photosynthesis	
Describe the function of the parts of a plant cell	
Give the word equation for photosynthesis	
Explain how plants are adapted for photosynthesis	
Describe the roles of different plant structures e.g. leaves, roots, stems, flowers	
Compare photosynthesis and respiration	



Knowledge Organiser
Year 7: Interdependence – Photosynthesis



A diagram showing the organelles in a typical plant cell

Animal vs. Plant Cells

What is cellular respiration?

The chemical word equation for aerobic respiration

What is photosynthesis?

The chemical word equation for photosynthesis

How does photosynthesis work?

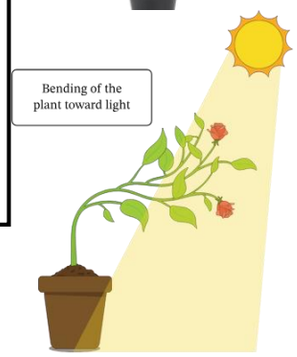
A diagram showing the reactants and products in photosynthesis

Adaptations for photosynthesis

Plants will try to create **a large surface area** to absorb the maximum amount of sun light



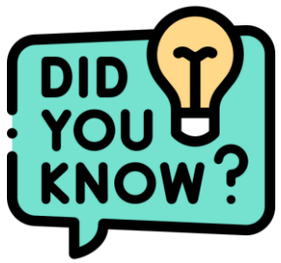






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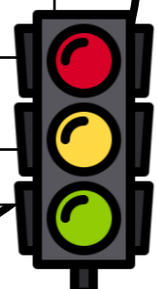
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